

ANN – Student Result Prediction

Python / TensorFlow Code

```
import pandas as pd
import tensorflow as tf
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler

data = {
    "study_hours": [
        1,2,2,3,3,3,4,4,4,5,
        5,5,6,6,6,7,7,7,8,8
    ],
    "attendance": [
        50,55,60,58,65,68,65,75,68,78,
        75,80,72,78,85,80,85,90,88,98
    ],
    "result": [
        0,0,0,0,0,0,1,1,1,1,
        1,1,1,1,1,1,1,1,1,1
    ],
}

df = pd.DataFrame(data)
x = df[["study_hours", "attendance"]]
y = df["result"]

X_train, X_test, y_train, y_test = train_test_split(
    x, y, test_size=0.25, random_state=42
)

scaler = StandardScaler()

X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)

model = tf.keras.Sequential([
    tf.keras.layers.Dense(8, activation="relu", input_shape=(2,)),
    tf.keras.layers.Dense(1, activation="sigmoid")
])

model.compile(
    optimizer="adam",
    loss="binary_crossentropy",
    metrics=["accuracy"]
)

model.fit(
    X_train,
    y_train,
    epochs=50,
    verbose=0
)

loss, accuracy = model.evaluate(X_test, y_test, verbose=0)

predictions = model.predict(X_test)

for actual, prediction in zip(y_test, predictions):
    predicted_class = 1 if prediction[0] >= 0.5 else 0

    print(
        "Actual:",
        actual,
        "Predicted:",
        predicted_class,
        "Probability:",
        predictions[0]
    )

print("Accuracy:", accuracy)
```